When KPIs Lie: Governance Signals for AI-Optimized Call Centers

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Abstract

Call center performance management systems increasingly rely on proxy metrics — particularly First Contact Resolution (FCR) — as primary indicators of service quality. This paper demonstrates that under operational pressure, rational agents systematically optimize proxy metrics in ways that diverge from durable customer outcomes, a phenomenon consistent with Goodhart's Law. Using a fully synthetic, open-source simulation environment (NovaWireless Call Center Lab), we generate 12 months of call center activity across 11 scenario types including metric gaming, fraudulent account manipulation, and loyalty offer failures. We show that proxy FCR remained stable near 89% while true durable resolution collapsed to approximately 47% — a 42 percentage-point Goodhart gap that was invisible to standard dashboards until Q4 churn exceeded forecast by over 11 percentage points. We introduce five governance signals — DAR, DOV, DRL, POR, and TER — that detect this divergence earlier and propose a governance framework for telecom executives seeking to align workforce incentives with long-run customer retention.

Keywords: Goodhart's Law, call center governance, first contact resolution, metric gaming, customer churn, AI workforce management, telecom operations

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Introduction

Modern call centers operate under intense performance pressure. Agents are evaluated on metrics that are easily measurable — handle time, call disposition codes, and First Contact Resolution (FCR) — because these metrics can be tracked in real time and rolled up into

executive dashboards. The assumption embedded in this practice is that these proxy metrics faithfully represent the underlying outcomes that matter: did the customer's problem actually get resolved? Will they call back? Will they stay?

This assumption is fragile. Goodhart's Law — the principle that when a measure becomes a target, it ceases to be a good measure — predicts that rational agents under pressure will find ways to satisfy the measure without satisfying the underlying goal. In call centers, this takes predictable forms: marking calls resolved before they are, applying small account credits to close escalations quickly, or avoiding difficult conversations that would extend handle time.

The consequences are not immediately visible. Proxy metrics look healthy. Dashboards stay green. Workforce planning, staffing models, and churn forecasts are all calibrated to a reality that does not exist. The gap accumulates silently until it surfaces — typically during a high-stress period such as Q4 holiday competition — as an unexplained surge in repeat contacts, customer complaints, and churn.

This paper makes three contributions. First, we document the anatomy of proxy metric gaming in telecom call centers through a fully synthetic simulation environment. Second, we introduce five governance signals — Delayed Adverse Rate (DAR), Durable Outcome Validation (DOV), Downstream Remediation Load (DRL), Proxy Overfit Ratio (POR), and Terminal Exit Rate (TER) — that detect Goodhart divergence before it reaches financial visibility. Third, we demonstrate the counterfactual: in a governance-aligned regime where reps are measured on honest outcomes rather than proxy metrics, the same workforce pressure produces burnout but not gaming.

Background and Related Work

Goodhart's Law in Performance Systems

Goodhart's Law was originally formulated in the context of monetary policy (Goodhart, 1975) but has since been applied broadly to organizational performance management (Muller, 2018). The core mechanism is simple: agents who are evaluated on a proxy metric have an incentive to optimize that metric directly, rather than the underlying construct it was designed to represent. This is rational individual behavior that produces irrational aggregate outcomes.

In customer service contexts, the problem is well-documented at the level of individual behavior — agents closing calls prematurely, inflating satisfaction scores through social pressure, or routing calls to avoid accountability. What is less studied is the system-level

consequence: how proxy metric optimization accumulates across a workforce over time, and what signals would allow a governance function to detect it before customer outcomes degrade visibly.

Call Center Metrics and Their Limitations

First Contact Resolution is widely regarded as the gold-standard call center metric because it proxies for both efficiency (no repeat contact) and quality (issue actually resolved). In practice, FCR is almost always measured as a proxy — either through agent self-report at call close, or through 30-day repeat contact windows that can miss delayed return calls. Neither approach can distinguish between a call that was genuinely resolved and one where the customer was temporarily satisfied by a credit that will expire, or where the underlying issue persists but the customer has not yet returned.

Average Handle Time (AHT) creates a secondary pressure. Reps who spend more time on difficult calls — the calls most likely to result in genuine resolution — are penalized in AHT dashboards relative to reps who close calls quickly through credits or deferrals. This creates a structural incentive to favor short, credit-based resolutions over longer, substantive ones.

The NovaWireless Simulation Environment

Design Philosophy

NovaWireless Call Center Lab is a fully open-source synthetic data generation environment designed to produce research-grade call center datasets that exhibit realistic Goodhart dynamics. The environment generates three interlinked datasets: a synthetic customer population with account histories, a call center workforce with individual performance profiles, and 12 months of call transcripts and outcome metadata.

The key design principle is that the simulation models both the proxy metric (what gets recorded in the CRM) and the true outcome (whether the customer's underlying problem was durably resolved). These diverge systematically across scenario types, allowing the Goodhart gap to be measured precisely — something that is impossible in real call center data, where ground truth resolution is unobservable.

Scenario Taxonomy

The simulation models 11 scenario types spanning the full range of call center dynamics. Table 1 summarizes the key parameters for each scenario.

Scenario	Proxy FCR	True FCR	Goodhart Gap	31-60d Repeat	Churn Mult.
clean	0.90	0.92	–0.02	0.04	1.00
unresolvable_clean	0.55	0.10	+0.45	0.25	1.40
gamed_metric	0.91	0.18	+0.73	0.45	1.25
fraud_store_promo	0.88	0.25	+0.63	0.40	1.50
fraud_line_add	0.85	0.22	+0.63	0.38	1.75
fraud_hic_exchange	0.87	0.15	+0.72	0.42	1.60
fraud_care_promo	0.92	0.30	+0.62	0.35	1.35
activation_clean	0.93	0.95	–0.02	0.03	0.90
activation_failed	0.45	0.08	+0.37	0.30	1.35
line_add_legitimate	0.90	0.92	–0.02	0.04	0.85
loyalty_offer_missed	0.93	0.14	+0.79	0.58	1.80

Table 1. Scenario taxonomy with proxy and true FCR, Goodhart gap, 31–60 day repeat contact probability, and churn multiplier relative to clean baseline.

The loyalty_offer_missed scenario deserves particular attention. It exhibits the highest proxy FCR of any gaming scenario (0.93) because the rep applies a small account credit and marks the call resolved. The underlying issue — a competitor offering superior device upgrade terms — is never addressed. The result is the highest 31-60 day repeat contact rate (0.58) and the highest churn multiplier (1.80) in the taxonomy, because the competitor offer remains on the table when the credit expires.

Workforce Pressure Experiment

To isolate the effect of workforce pressure on metric gaming, we ran a controlled experiment comparing a baseline condition (base strain = 0.52, pressure = 0.14) against a high-pressure condition (base strain = 0.72, pressure = 0.62). Each condition synthesized 250 representative agents and simulated 5,000 calls.

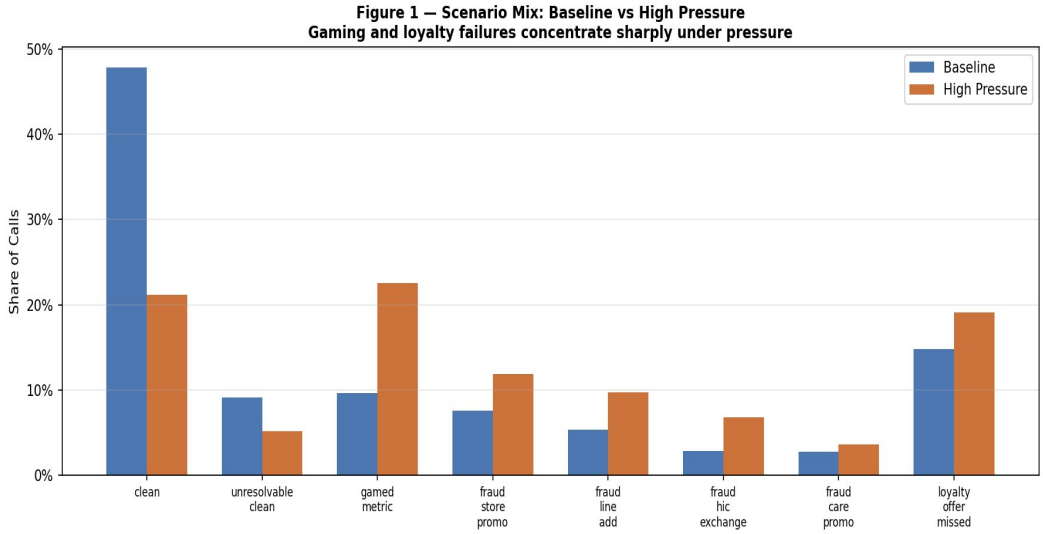

Figure 1. Scenario mix shift under high pressure. Clean calls decrease from 48% to 21% while gaming scenarios and loyalty failures concentrate. Gaming and loyalty_offer_missed together account for 42% of calls under high pressure versus 25% at baseline.

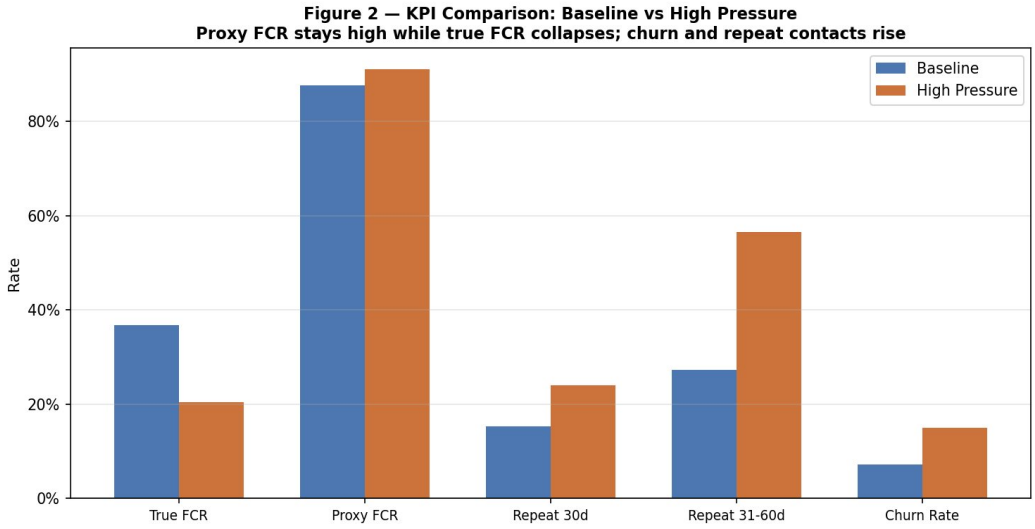

Figure 2. KPI comparison across conditions. Proxy FCR remains elevated (89% baseline, 91% high pressure) while true FCR declines (37% to 21%). Repeat 31-60d contacts nearly double and churn rate doubles under high pressure.

The Goodhart Gap: Measurement and Anatomy

Aggregate Gap Dynamics

Across the full 12-month simulation, the aggregate proxy FCR held near 89% while true durable FCR was approximately 47% — a gap of 42 percentage points. This gap is not

uniformly distributed across time. It begins at approximately 47 percentage points in January and widens progressively through the year as the scenario mix shifts toward higher-gap scenarios, reaching 71.6 percentage points in December.

The widening gap is a direct consequence of the loyalty_offer_missed scenario scaling with competitor promotional intensity. As competitor holiday promotions intensify through Q3 and Q4, the share of calls driven by competitive comparison increases. Reps who lack awareness of the trade-in loyalty upgrade program — or who lack the time under pressure to navigate it — apply band-aid credits and close the call. The dashboard records a resolved call. The customer is temporarily satisfied. Thirty to sixty days later, when the credit expires and the competitor offer is still active, the customer calls back or churns.

Per-Scenario Gap Structure

Figure 4 shows the per-scenario Goodhart gap under both conditions. The loyalty_offer_missed scenario shows the largest gap at baseline (approximately 79 percentage points) and widens further under high pressure. The gamed_metric scenario shows a large gap (approximately 79–84 percentage points) but with a different mechanism: here the rep is actively gaming the CRM disposition rather than applying a legitimate but insufficient resolution.

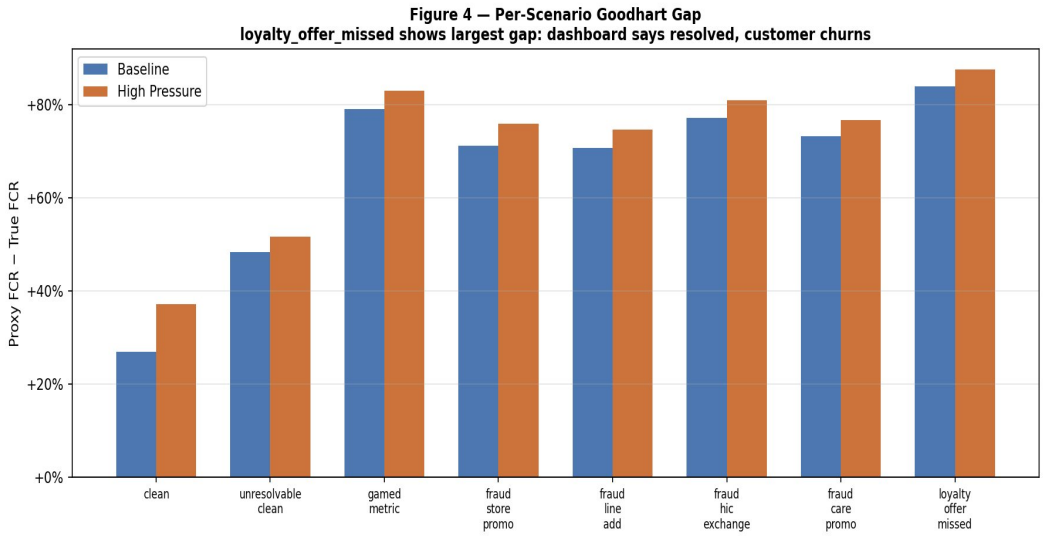

Figure 4. Per-scenario Goodhart gap (Proxy FCR minus True FCR) under baseline and high-pressure conditions. loyalty_offer_missed shows the largest gap, reflecting high proxy FCR driven by credit application against low true resolution of the underlying competitive churn risk.

Workforce Strain and Gaming Propensity

Figure 3 shows the distribution of burnout and strain scores across the two workforce conditions. The high-pressure condition shifts the entire population rightward — not just the tail. Under baseline conditions, the majority of reps fall in the low-to-medium strain range. Under high pressure, the distribution centers near the high-strain threshold. This matters because gaming propensity in the simulation is a function of both burnout and compliance risk: reps with high burnout and high compliance risk are significantly more likely to route toward gaming scenarios when given the choice.

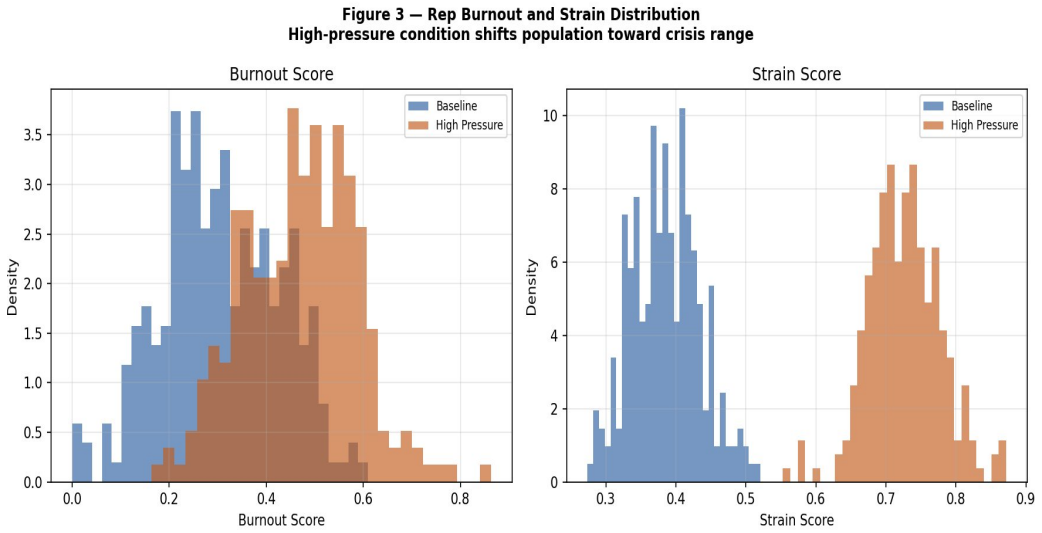

Figure 3. Rep burnout and strain distribution across conditions. The high-pressure condition shifts the population into crisis range, increasing the proportion of reps with elevated gaming propensity.

The Q4 Breaking Point

Figure 5 shows the three signals that appeared too late. The top panel shows that proxy FCR (the dashboard metric) remained stable and slightly improving throughout the year — reaching 91% in December — while true durable FCR declined steadily from 40% in January to 20% in December. The gap reached 71.6 percentage points by Q4.

The middle panel shows repeat contact dynamics. Thirty-day repeat contacts grew modestly, from approximately 12% in January to 29% in December. This would register as a manageable trend in most operations reviews. The 31-60 day repeat contacts — the signal of band-aid resolutions expiring — grew dramatically, from approximately 21% in January to 60% in December. This is the signature of the loyalty_offer_missed scenario at scale: customers who were temporarily satisfied returning en masse when their credits expire.

The bottom panel shows the financial consequence. The company's churn forecast, calibrated to proxy FCR, held near 4.2% throughout the year. Actual churn climbed from approximately 6.8% in January — already above forecast — to 16% in December, an excess of approximately 11.8 percentage points above the forecast. This gap between forecast and actual represents the staffing, retention budget, and revenue planning errors that accumulate when the measurement system is compromised.

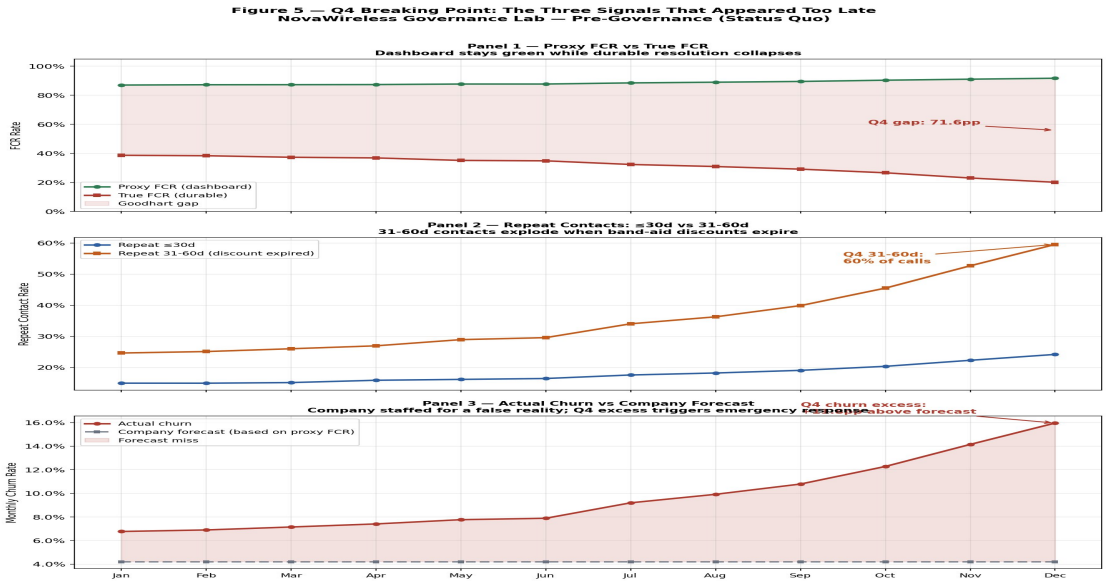

Figure 5. The three Q4 signals. Panel 1: Proxy FCR stays near 90% while true FCR declines to 20%, reaching a 71.6pp gap. Panel 2: 31-60 day repeat contacts reach 60% in December as band-aid credits expire. Panel 3: Actual churn reaches 16% against a 4.2% forecast, an excess of 11.8pp that triggers emergency response.

Governance Signals for Early Detection

The five governance signals we propose are designed to detect Goodhart divergence at different stages of the causal chain, giving operations leaders multiple opportunities to intervene before the divergence reaches financial visibility.

DAR — Delayed Adverse Rate

DAR measures the rate of adverse customer outcomes (repeat contacts, escalations, churn) that occur 31-60 days after a call was marked resolved. A high DAR indicates that resolutions recorded at call close are not durable — the issue is returning after a delay consistent with a temporary credit or deferral. DAR is the most direct leading indicator of the loyalty_offer_missed and gamed_metric scenarios.

DOV — Durable Outcome Validation

DOV measures the correlation between proxy resolution (CRM disposition) and true durable resolution (no adverse outcome within 60 days) at the rep and scenario level. A declining DOV score indicates that the mapping between what reps record and what customers experience is degrading. DOV can be computed retrospectively from existing data and does not require changes to CRM instrumentation.

DRL — Downstream Remediation Load

DRL measures the volume of downstream work generated by calls that were previously marked resolved — escalations, credits, retention interventions, and churn processing that trace back to calls in the 31-60 day window. High DRL indicates that the resolution recorded at call close is generating future costs that are not being attributed to the original interaction.

POR — Proxy Overfit Ratio

POR measures the ratio of proxy FCR improvement to true FCR improvement over a rolling window. A POR greater than 1.0 indicates that the proxy metric is improving faster than the underlying outcome — a direct signal of gaming. POR above 2.0 should trigger immediate governance review. In the NovaWireless simulation, POR reached approximately 3.6 by Q4 as proxy FCR continued improving while true FCR declined.

TER — Terminal Exit Rate

TER measures the share of customers who churn within 90 days of a call that was marked resolved with a credit or concession. High TER indicates that credit-based resolutions are not preventing churn — they are delaying it. TER is particularly sensitive to the loyalty_offer_missed scenario because the underlying competitive offer remains active after the credit expires.

The Governance Regime: A Counterfactual

A central claim of this paper is that the problem is not pressure — it is what pressure is applied to. In the governance-aligned simulation, reps are evaluated on DAR, DOV, and trust delta rather than proxy FCR. There is no metric to game. The only pathway to gaming behavior under this regime is pure burnout: a rep so overwhelmed that they cut corners not because they are chasing a number, but because they are breaking down.

Under the governance regime, the same high-pressure conditions produce elevated burnout and compliance risk — these are real consequences of understaffing and cannot be eliminated by changing the measurement system. What is eliminated is the systematic scenario mix shift: gaming scenario concentration does not increase under pressure when there is no proxy metric to inflate. The Goodhart gap remains small. Repeat contacts and churn track close to their true rates. The dashboard reflects reality.

This counterfactual is not a claim that governance eliminates operational problems. It is a claim that governance determines whether operational problems are visible or hidden. Under proxy metric regimes, pressure hides problems. Under governance regimes, pressure surfaces them — which is exactly what an operations leader needs in order to act.

Implications for Telecom Executives

What to Stop Doing

First Contact Resolution, as currently measured — through agent self-report or 30-day repeat contact windows — should not be used as a primary performance metric for individual agents. It is too easily gamed and the gaming is too closely aligned with rational agent behavior under pressure to be detectable through supervision alone. Average Handle Time targets that reward speed over thoroughness should similarly be restructured or eliminated.

What to Implement

The five governance signals described above can be implemented using data already available in most telecom CRM and billing systems. DAR and TER require linking call records to outcome events at the 31-60 day and 90-day windows — a straightforward join that most data engineering teams can implement in days. DOV requires a retrospective scoring pass over existing call records. POR and DRL can be computed as rolling aggregates on any modern analytics platform.

The governance framework does not require eliminating performance measurement — it requires replacing proxy metrics with outcome metrics that cannot be gamed through call disposition behavior. Reps measured on DAR and DOV are incentivized to achieve durable resolutions, because gaming the metric would require actually fixing the customer's problem.

The Loyalty Offer Gap

The loyalty_offer_missed scenario represents a structural product gap, not a rep performance problem. Reps who are unaware of the trade-in loyalty upgrade program, or who lack the system access or time to navigate it under pressure, will continue applying band-aid credits regardless of the measurement regime. The governance signal (high DAR on competitive comparison calls) identifies the problem; the solution is a product and training intervention, not a performance management intervention.

Conclusion

This paper has documented the anatomy of proxy metric gaming in telecom call centers and demonstrated that the consequences — a 42 percentage-point Goodhart gap, 60% repeat contact rates, and 12 percentage-point churn forecast errors — are predictable, detectable, and preventable.

The fundamental insight is that measurement systems shape behavior, and behavior shapes outcomes. When the measurement system rewards the appearance of resolution over actual resolution, rational agents produce the appearance of resolution. The dashboard stays green. The customer churns. The company plans for a future that does not exist.

The governance signals introduced here — DAR, DOV, DRL, POR, and TER — do not require new data collection infrastructure. They require a commitment to measuring outcomes rather than proxies, and to building accountability systems that make gaming costly rather than rewarded. The simulation environment is open source and can be adapted to any telecom operating context.

The Q4 breaking point is not inevitable. It is the predictable consequence of a measurement system that cannot see what it is measuring. The signals were there. They simply were not being watched.

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